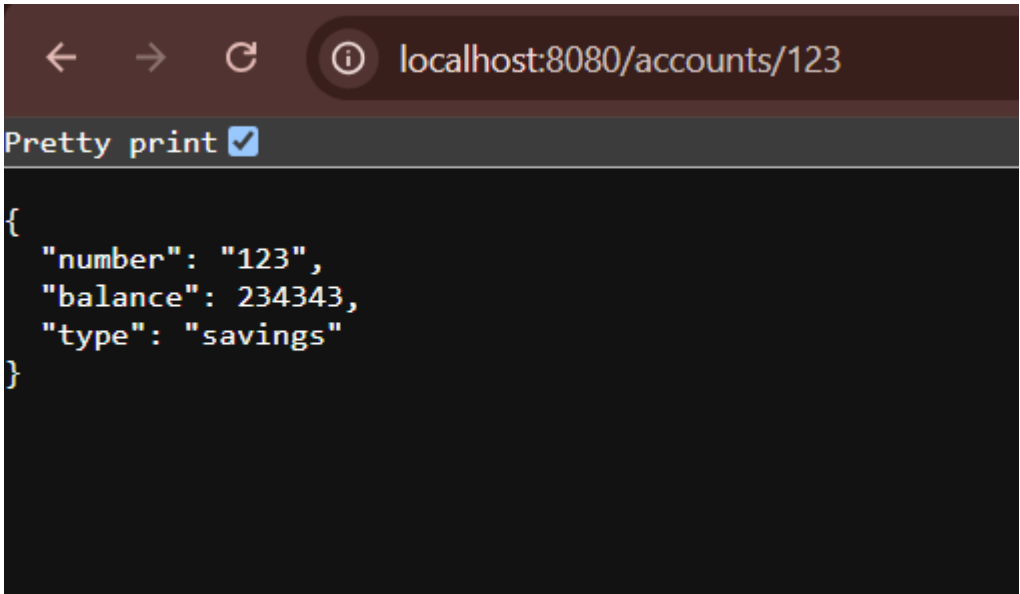
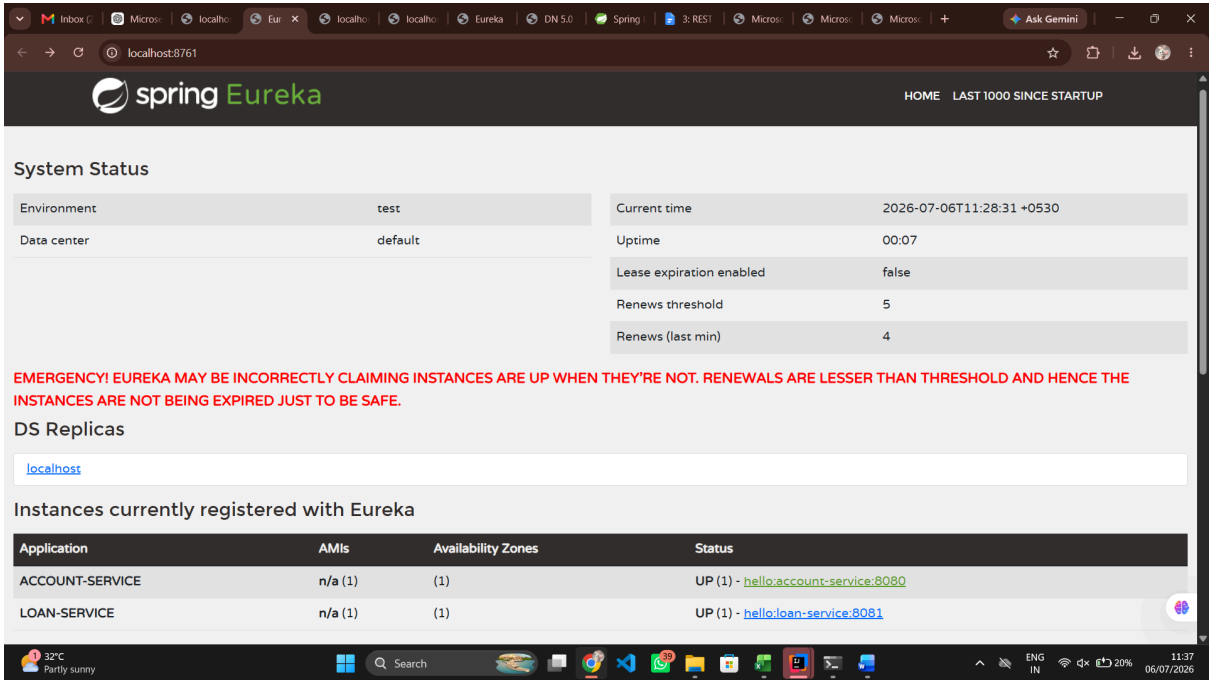
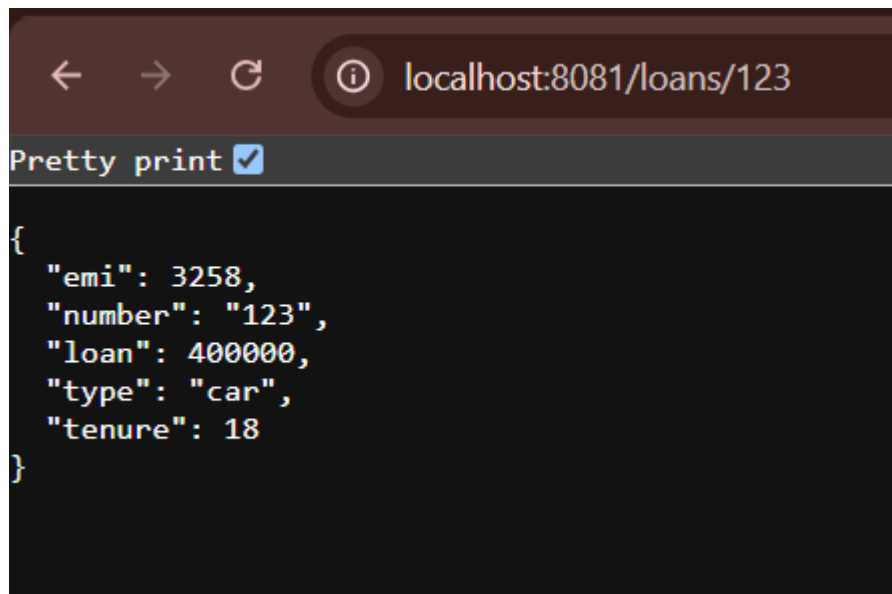


Output:

Output:
Before setting gateway



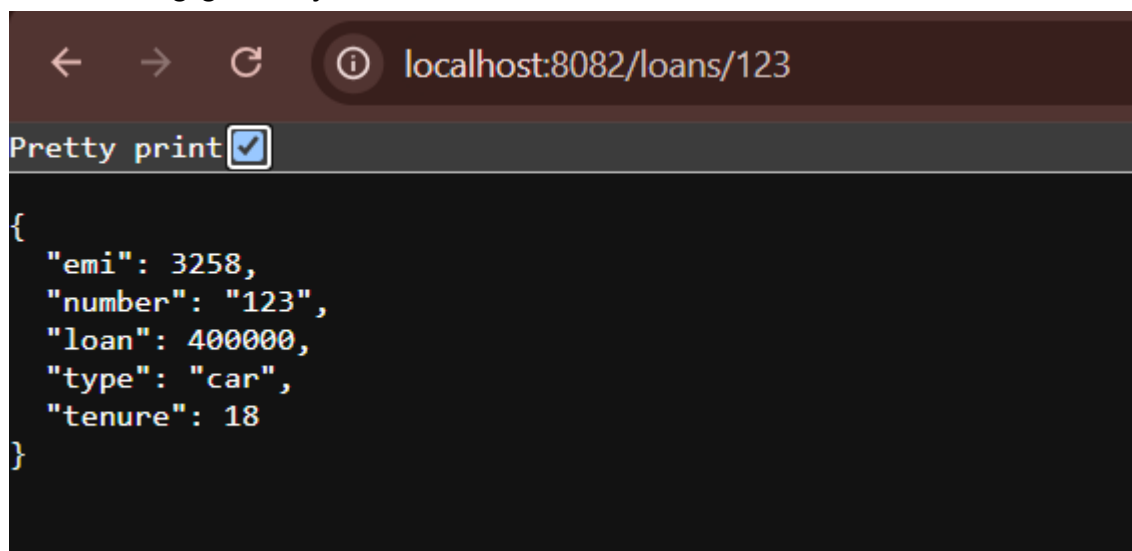


```
localhost:8081/loans/123

Pretty print ☒

{
  "emi": 3258,
  "number": "123",
  "loan": 400000,
  "type": "car",
  "tenure": 18
}
```

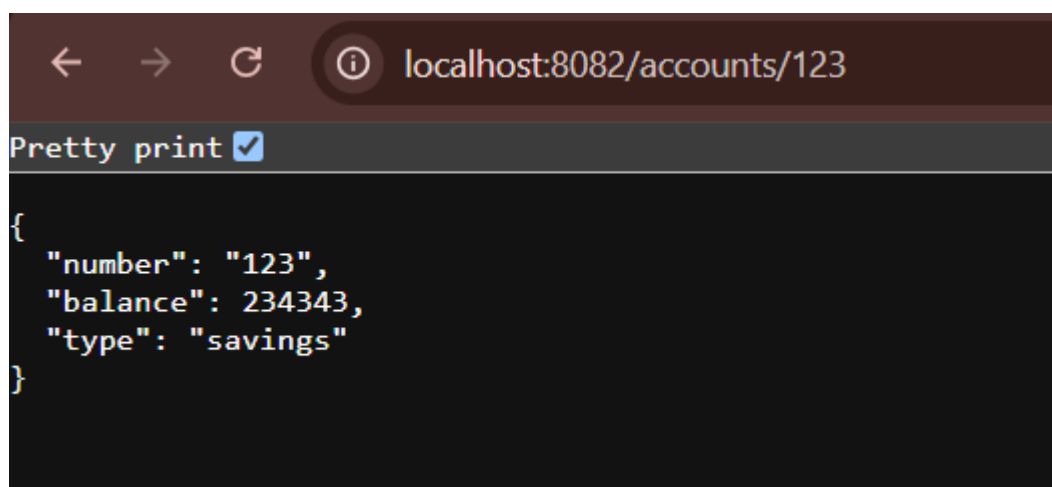
After setting gateway



```
localhost:8082/loans/123

Pretty print ☒

{
  "emi": 3258,
  "number": "123",
  "loan": 400000,
  "type": "car",
  "tenure": 18
}
```



```
localhost:8082/accounts/123

Pretty print ☒

{
  "number": "123",
  "balance": 234343,
  "type": "savings"
}
```

The screenshot shows the Spring Eureka web interface. The top navigation bar includes 'HOME' and 'LAST 1000 SINCE STARTUP'. The 'System Status' section displays the following information:

Environment	test	Current time	2026-07-06T11:53:18 +0530
Data center	default	Uptime	00:32
		Lease expiration enabled	true
		Renews threshold	6
		Renews (last min)	12

The 'DS Replicas' section shows a single replica at 'localhost'.

The 'Instances currently registered with Eureka' section displays a table of registered instances:

Application	AMIs	Availability Zones	Status
ACCOUNT-SERVICE	n/a (1)	(1)	UP (1) - hello:account-service:8080
API-GATEWAY	n/a (1)	(1)	UP (1) - hello:api-gateway:8082
LOAN-SERVICE	n/a (1)	(1)	UP (1) - hello:loan-service:8081

The 'General Info' section shows the current weather as 33°C Partly sunny.

EUREKA

application.properties

The screenshot shows an IDE with the 'application.properties' file open. The file contains the following properties:

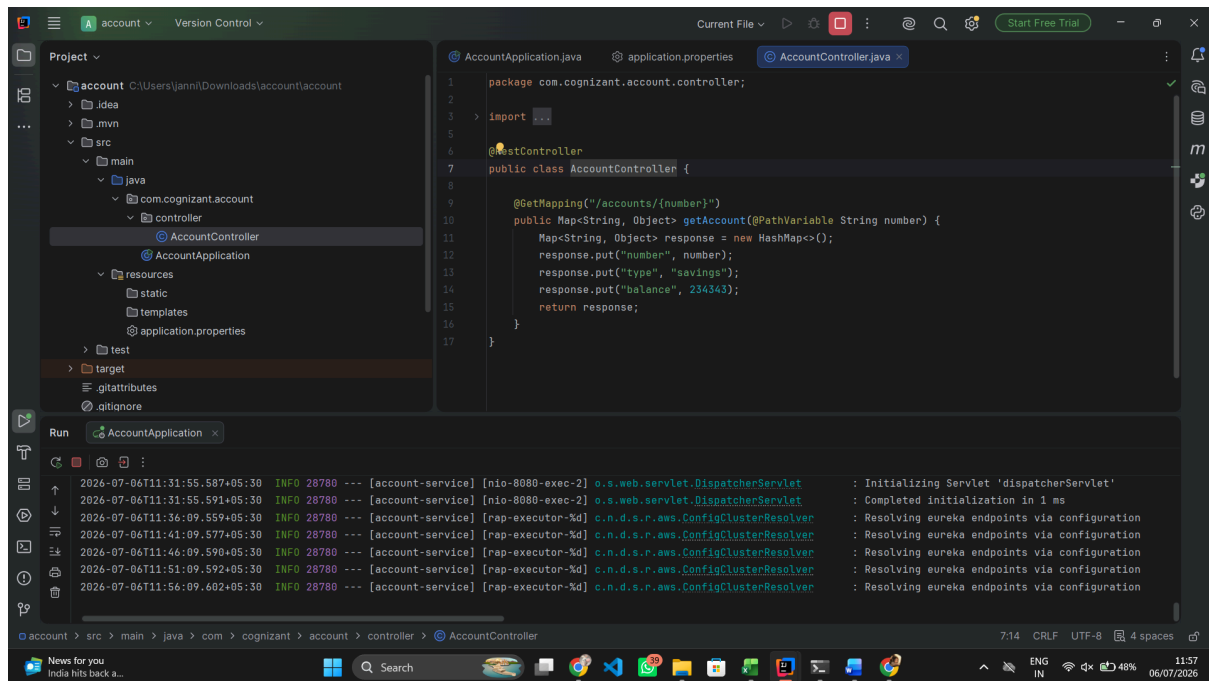
```
spring.application.name=eureka-discovery-server
server.port=8761
eureka.client.register-with-eureka=false
eureka.client.fetch-registry=false
```

The console output shows the following logs:

```
2026-07-06T11:52:28.750+05:30 WARN 14256 --- [eureka-discovery-server] [get_localhost-5] c.n.eureka.cluster.ReplicationTask : The replication of task API-GATEWAY/hello:
2026-07-06T11:52:28.750+05:30 WARN 14256 --- [eureka-discovery-server] [get_localhost-5] c.netflix.eureka.cluster.PeerEurekaNode : API-GATEWAY/hello:api-gateway:8082:Cancel@
2026-07-06T11:52:32.839+05:30 INFO 14256 --- [eureka-discovery-server] [nio-8761-exec-10] c.n.e.registry.AbstractInstanceRegistry : Registered instance API-GATEWAY/hello:api-
2026-07-06T11:52:33.359+05:30 INFO 14256 --- [eureka-discovery-server] [nio-8761-exec-5] c.n.e.registry.AbstractInstanceRegistry : Registered instance API-GATEWAY/hello:api-
2026-07-06T11:52:56.128+05:30 INFO 14256 --- [eureka-discovery-server] [a-EvictionTimer] c.n.e.registry.AbstractInstanceRegistry : Running the evict task with compensationTi
2026-07-06T11:53:56.129+05:30 INFO 14256 --- [eureka-discovery-server] [a-EvictionTimer] c.n.e.registry.AbstractInstanceRegistry : Running the evict task with compensationTi
2026-07-06T11:54:56.131+05:30 INFO 14256 --- [eureka-discovery-server] [a-EvictionTimer] c.n.e.registry.AbstractInstanceRegistry : Running the evict task with compensationTi
```

Main -> add this `@EnableEurekaServer`

ACCOUNT: accountController

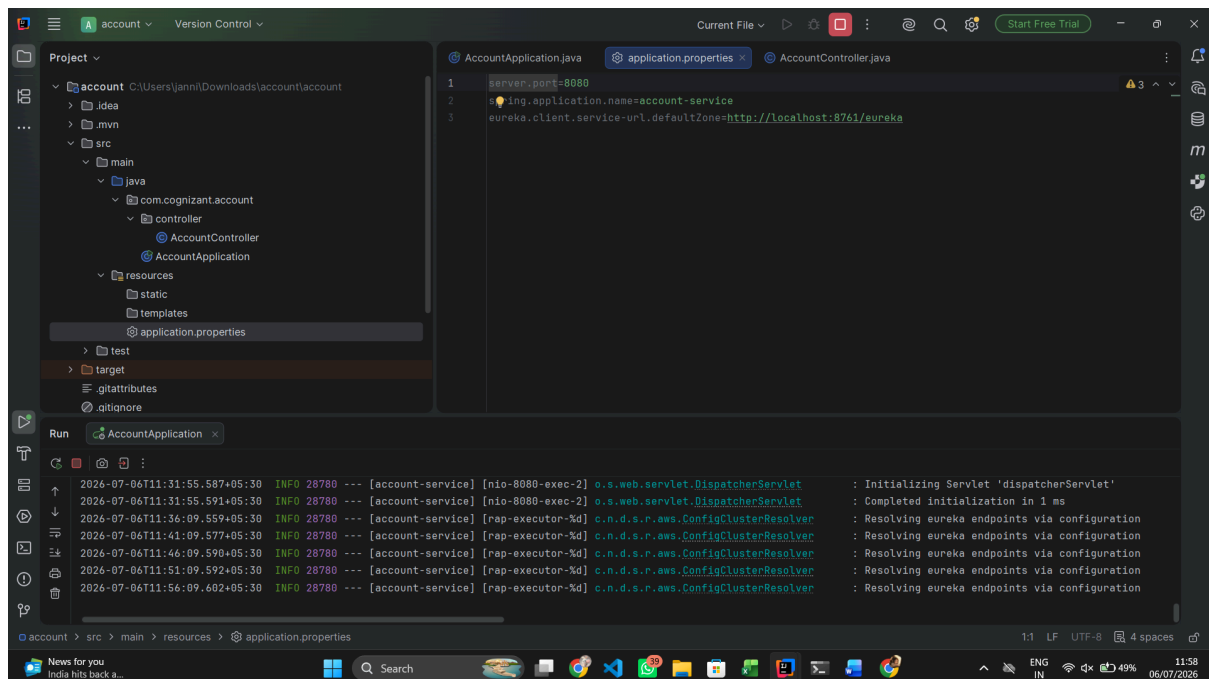


```
1 package com.cognizant.account.controller;
2
3 import ...
4
5 @RestController
6 public class AccountController {
7
8     @GetMapping("/accounts/{number}")
9     public Map<String, Object> getAccount(@PathVariable String number) {
10         Map<String, Object> response = new HashMap<>();
11         response.put("number", number);
12         response.put("type", "savings");
13         response.put("balance", 234343);
14         return response;
15     }
16 }
17 }
```

Run console output:

```
2026-07-06T11:31:55.587+05:30 INFO 28780 --- [account-service] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-06T11:31:55.591+05:30 INFO 28780 --- [account-service] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-07-06T11:36:09.559+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-06T11:41:09.577+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-06T11:46:09.590+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-06T11:51:09.592+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-06T11:56:09.602+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
```

application.properties



```
1 server.port=8080
2 spring.application.name=account-service
3 eureka.client.service-url.defaultZone=http://localhost:8761/eureka
```

Run console output:

```
2026-07-06T11:31:55.587+05:30 INFO 28780 --- [account-service] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-06T11:31:55.591+05:30 INFO 28780 --- [account-service] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-07-06T11:36:09.559+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-06T11:41:09.577+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-06T11:46:09.590+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-06T11:51:09.592+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-06T11:56:09.602+05:30 INFO 28780 --- [account-service] [rap-executor-8d] c.n.d.s.p.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
```

main -> add `@EnableDiscoveryClient`

LOAN

loanController

```
8 public class LoanController {
9
10     @GetMapping("/loans/{number}")
11     public Map<String, Object> getLoan(@PathVariable String number) {
12
13         Map<String, Object> response = new HashMap<>();
14         response.put("number", number);
15         response.put("type", "car");
16         response.put("loan", 400000);
17         response.put("emi", 3250);
18         response.put("tenure", 10);
19
20         return response;
21     }
22 }
```

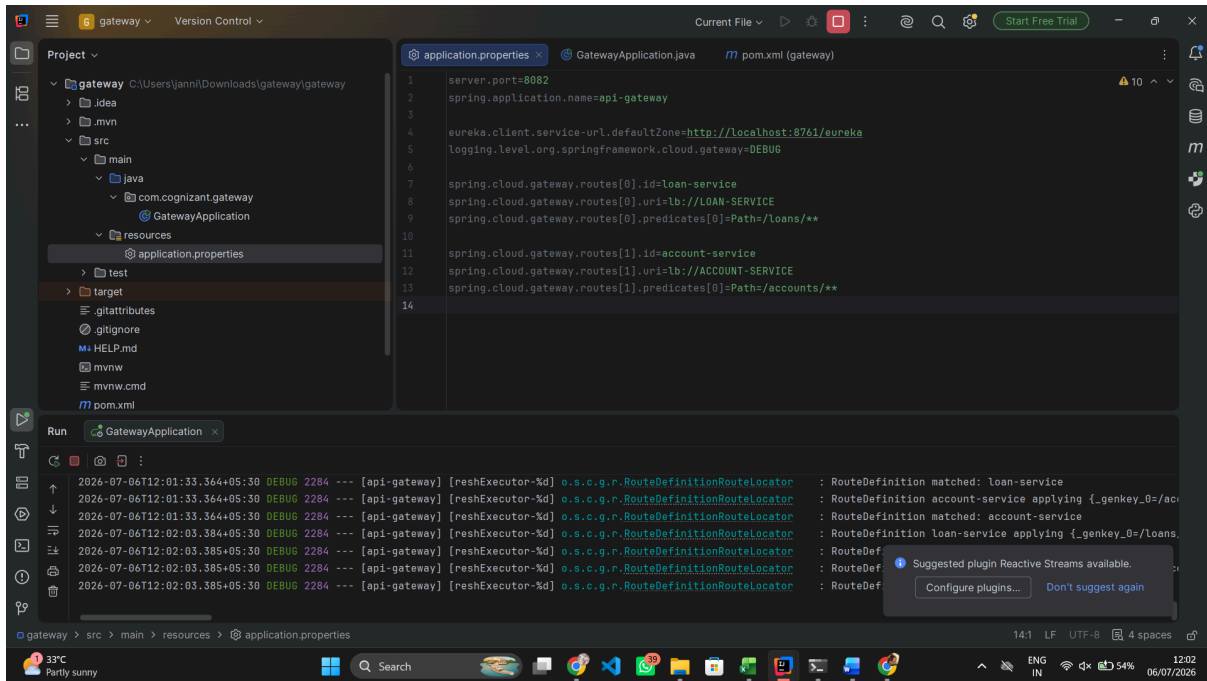
application.properties

```
1 server.port=8081
2 spring.application.name=loan-service
3 eureka.client.service-url.defaultZone=http://localhost:8761/eureka
```

main -> add @EnableDiscoveryClient

GATEWAY

application.properties



The screenshot displays an IDE interface with the following components:

- Project Explorer:** Shows the project structure with folders like `idea`, `mvn`, `src`, `main`, `java`, `com.cognizant.gateway`, `resources`, `test`, `target`, `gitattributes`, `.gitignore`, `HELP.md`, `mvnw`, `mvnw.cmd`, and `pom.xml`. The `application.properties` file is selected under the `resources` folder.
- Editor:** Displays the content of `application.properties` with the following configuration:

```
1 server.port=8082
2 spring.application.name=api-gateway
3
4 eureka.client.service-url.defaultZone=http://localhost:8761/eureka
5 logging.level.org.springframework.cloud.gateway=DEBUG
6
7 spring.cloud.gateway.routes[0].id=loan-service
8 spring.cloud.gateway.routes[0].uri=lb://LOAN-SERVICE
9 spring.cloud.gateway.routes[0].predicates[0]=Path=/loans/**
10
11 spring.cloud.gateway.routes[1].id=account-service
12 spring.cloud.gateway.routes[1].uri=lb://ACCOUNT-SERVICE
13 spring.cloud.gateway.routes[1].predicates[0]=Path=/accounts/**
14
```
- Run Console:** Shows the output of the `GatewayApplication` run. The log messages indicate that the routes are being matched and applied:

```
2026-07-06T12:01:33.364+05:30 DEBUG 2284 --- [api-gateway] [reshExecutor-5d] o.s.c.g.r.RouteDefinitionRouteLocator : RouteDefinition matched: loan-service
2026-07-06T12:01:33.364+05:30 DEBUG 2284 --- [api-gateway] [reshExecutor-5d] o.s.c.g.r.RouteDefinitionRouteLocator : RouteDefinition account-service applying {_genkey_0=/accounts/**}
2026-07-06T12:01:33.364+05:30 DEBUG 2284 --- [api-gateway] [reshExecutor-5d] o.s.c.g.r.RouteDefinitionRouteLocator : RouteDefinition matched: account-service
2026-07-06T12:01:33.364+05:30 DEBUG 2284 --- [api-gateway] [reshExecutor-5d] o.s.c.g.r.RouteDefinitionRouteLocator : RouteDefinition loan-service applying {_genkey_0=/loans/**}
2026-07-06T12:02:03.385+05:30 DEBUG 2284 --- [api-gateway] [reshExecutor-5d] o.s.c.g.r.RouteDefinitionRouteLocator : RouteDefinition matched: loan-service
2026-07-06T12:02:03.385+05:30 DEBUG 2284 --- [api-gateway] [reshExecutor-5d] o.s.c.g.r.RouteDefinitionRouteLocator : RouteDefinition account-service applying {_genkey_0=/accounts/**}
2026-07-06T12:02:03.385+05:30 DEBUG 2284 --- [api-gateway] [reshExecutor-5d] o.s.c.g.r.RouteDefinitionRouteLocator : RouteDefinition matched: account-service
2026-07-06T12:02:03.385+05:30 DEBUG 2284 --- [api-gateway] [reshExecutor-5d] o.s.c.g.r.RouteDefinitionRouteLocator : RouteDefinition loan-service applying {_genkey_0=/loans/**}
```
- Notification:** A message box suggests the `Reactive Streams` plugin is available, with buttons for `Configure plugins...` and `Don't suggest again`.